

Research Proposal

Curiosity-Driven Reinforcement Learning for Emergent Tool Use on a Humanoid Robot

Proposed Platform: Unitree G1 EDU Humanoid

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Abstract

We propose a research project investigating whether a full-sized humanoid robot can discover tool use behaviors purely through curiosity-driven reinforcement learning, without demonstrations, pre-programmed heuristics, or shaped rewards for tool interaction. Using the Unitree G1 EDU humanoid as our target platform, we will construct sparse-reward environments in NVIDIA Isaac Sim where task goals are deliberately unreachable without creative use of nearby objects — for example, pressing an elevated button by standing on a box, or retrieving an occluded item by using a stick to extend reach. An intrinsic curiosity module (Random Network Distillation) will drive exploration, while an asymmetric actor-critic architecture will bridge simulation and real-world deployment. If successful, this work will constitute the first demonstration of emergent tool use on a bipedal humanoid, combining whole-body locomotion, dexterous manipulation, and open-ended problem solving within a single learned policy.

1. Introduction & Motivation

Tool use has long been considered a hallmark of higher intelligence in both biological and artificial agents. In animal cognition, the capacity to repurpose environmental objects to achieve otherwise impossible goals — crows bending wire to hook food, chimpanzees using sticks to extract termites — is treated as strong evidence of flexible reasoning and planning. In robotics, however, tool use remains almost exclusively scripted: a robot arm is given a known tool, a known grasp pose, and a known task, and executes a pre-planned trajectory.

Emergent tool use — where the agent independently discovers that an environmental object can serve as a tool to solve an otherwise intractable problem — is a fundamentally different and far more challenging capability. It requires the agent to jointly reason about its own body, the physics of candidate objects, and the structure of the task, all without explicit instruction. In the reinforcement learning community, emergent tool use has been demonstrated in simplified 2D or tabletop simulation environments (e.g., OpenAI's multi-agent hide-and-seek, where agents learned to use ramps and boxes to overcome barriers). However, no prior work has demonstrated emergent tool use on a full-sized humanoid robot with a realistic morphology, dexterous hands, and bipedal locomotion constraints.

We believe the Unitree G1 EDU platform — with its 43 degrees of freedom, Dex3-1 force-controlled hands, onboard NVIDIA Jetson Orin (275 TOPS), and comprehensive RL toolchain (unitree_rl_gym, Isaac Sim URDF) — presents a unique opportunity to attempt this. The G1's physical capabilities are sufficient for the manipulation and locomotion primitives that tool use demands, and its simulation fidelity in Isaac Lab / MuJoCo has been validated by recent work such as OmniXtreme (BIGAI & Unitree, Feb 2026) and VideoMimic (2025).

2. Research Questions

This project is organized around three core research questions:

- **RQ1 — Emergence:** Can a humanoid agent discover tool-use behaviors (e.g., using a box as a step stool, a stick as a reach extender) in sparse-reward environments purely through intrinsic curiosity, without any demonstrations or tool-specific reward shaping?
- **RQ2 — Whole-Body Coordination:** Does the humanoid morphology (bipedal balance, dexterous manipulation, large workspace) enable qualitatively different tool-use strategies compared to fixed-base manipulators — for instance, carrying a tool while walking, or repositioning the body to create leverage?
- **RQ3 — Transferability:** Can policies trained with curiosity-driven RL in simulation transfer to the physical G1 via standard sim-to-real techniques (domain randomization, asymmetric actor-critic), and do the emergent tool-use behaviors survive the reality gap?

3. Related Work

3.1 Emergent Tool Use in RL

Baker et al. (OpenAI, 2020) demonstrated emergent tool use in a multi-agent hide-and-seek game, where agents learned to use ramps, boxes, and walls to gain advantages over opponents. However, this was in a simplified 2D/3D environment with point-mass-like agents, not embodied robots. Subsequent work by Berner et al. explored emergent strategies in Dota 2 but without physical grounding. In robotic manipulation,

Fang et al. (2020) and Xu et al. (2023) studied tool use with robot arms, but these approaches relied on task-specific reward shaping or demonstrations of tool grasp and use, removing the emergent discovery component.

3.2 Curiosity-Driven Exploration

Intrinsic motivation methods — particularly Random Network Distillation (RND, Burda et al. 2019) and the Intrinsic Curiosity Module (ICM, Pathak et al. 2017) — have proven effective at driving exploration in sparse-reward environments. RND provides a prediction-error-based curiosity signal that decays as the agent visits familiar states, naturally guiding it toward novel interactions. These methods have been applied to Atari games and simple continuous control but not to humanoid loco-manipulation in photorealistic physics simulators.

3.3 Humanoid RL on the Unitree G1

The G1 has become a standard RL research platform. OmniXtreme (Wang et al., Feb 2026) achieved scalable high-dynamic motion tracking using flow-matching policies with residual RL refinement. VideoMimic (2025) demonstrated context-aware locomotion learned from monocular human videos. Unitree’s own `unitree_rl_gym` provides PPO-based locomotion training with sim-to-real transfer. FAME (2026) and FALCON (2025) explored force-adaptive loco-manipulation. However, all existing G1 RL work focuses on tracking known motions or executing pre-defined manipulation tasks — none addresses open-ended, curiosity-driven discovery of novel behaviors.

4. Proposed Approach & Pipeline

Our system is organized into four stages: environment design, curiosity-driven training, policy distillation, and sim-to-real transfer. The figure below illustrates the end-to-end pipeline.

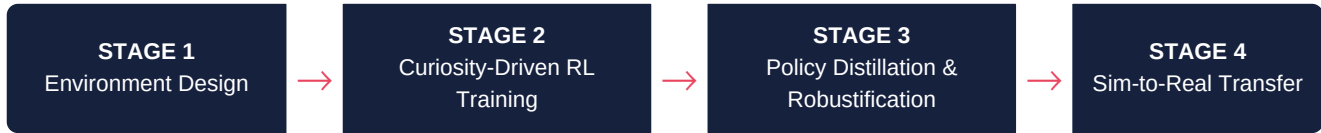


Figure 1. End-to-end research pipeline overview.

4.1 Stage 1 — Environment Design (Isaac Lab)

We will design a suite of task environments in NVIDIA Isaac Lab, each structured so that the goal state is physically unreachable using only the G1's body. The environments share a common structure: a goal (button press, object retrieval, surface reach), a set of "distractor" objects with varied physical properties (mass, friction, shape), and one or more objects that can function as tools if the agent discovers how.

Proposed task suite:

Task	Goal	Latent Tool Solution	Difficulty
Elevated Button	Press a button mounted 2.0 m high (G1 reach $\approx$ 1.6 m)	Push a box underneath, climb onto it, press button	Medium
Occluded Retrieval	Retrieve object behind a narrow gap/barrier	Use a stick/rod to push or hook the object out	Medium
Distant Target	Touch a target beyond arm's reach across a gap	Pick up a long stick to bridge the reach gap	Easy
Weight Lever	Lift a heavy object beyond G1's payload	Use a plank as a lever with a fulcrum object	Hard
Composite	Navigate obstacle, then reach elevated goal	Sequence: carry box $\rightarrow$ place $\rightarrow$ climb $\rightarrow$ reach	Hard

Table 1. Proposed task environments with latent tool-use solutions.

All environments use the same G1 EDU URDF (43 DOF, Dex3-1 hands). Objects are procedurally randomized in position, orientation, mass ($\pm 30\%$), friction ($\pm 50\%$), and color to prevent overfitting. The extrinsic reward is binary and sparse: +1 only upon goal completion, 0 otherwise. A small per-timestep penalty (-0.001) encourages efficiency but does not guide strategy.

4.2 Stage 2 — Curiosity-Driven RL Training

The core of our approach replaces dense, hand-crafted reward shaping with an intrinsic curiosity signal. We adopt Random Network Distillation (RND) as the primary curiosity mechanism, with the option to benchmark against ICM.

Architecture — Asymmetric Actor-Critic with RND:

- **Actor (deployed policy):** Receives ego-centric observations only — proprioception (joint positions, velocities, torques), IMU (orientation, angular velocity), wrist force-torque, and a 64×64 egocentric depth image from the G1's head camera. Outputs target joint positions at 50 Hz. This is the policy that transfers to real hardware.
- **Critic (training only):** Receives privileged information — full object poses, contact states, goal position, terrain mesh — enabling faster value estimation without leaking privileged features into the deployable actor.
- **RND module:** A fixed random neural network and a trainable predictor network both take the actor's observation as input. The prediction error serves as the intrinsic reward signal $r_{\text{intrinsic}} = \|f_{\text{RND}}(o) - f(o)\|^2$. As the agent revisits known states, the predictor improves and curiosity decays, pushing the agent to explore novel object interactions.

The total reward is: $r_{\text{total}} = r_{\text{extrinsic}} + \alpha \cdot r_{\text{intrinsic}}$, where α is annealed from 1.0 $\rightarrow$ 0.1 over training to shift from exploration to exploitation as the agent discovers successful strategies.

Training details:

- Algorithm: PPO (clipped surrogate objective) with GAE ($\lambda = 0.95$)
- Parallel environments: 4,096 instances (Isaac Lab GPU parallelism)
- Curriculum: Environments are staged by difficulty — distant target $\rightarrow$ elevated button $\rightarrow$ occluded retrieval $\rightarrow$ composite tasks. Each environment unlocks when the agent achieves >20% success rate on the current one.
- Observation normalization: Running mean/variance normalization on all observation channels
- Episode horizon: 2,000 steps (40 seconds at 50 Hz) per episode

4.3 Stage 3 — Policy Distillation & Robustification

Once the curiosity-driven policy discovers tool-use behaviors, we apply a two-phase refinement process inspired by OmniXtreme's architecture:

- **Phase A — Behavior Cloning Distillation:** The best-performing curiosity policy rollouts are collected as demonstration data. A clean student policy (same architecture, no RND overhead) is trained via DAgger to reproduce the discovered behaviors. This removes the exploration noise and produces a compact, deployable network.
- **Phase B — Residual RL Refinement:** The distilled base policy is frozen. A lightweight residual MLP is trained on top using RL with realistic actuator constraints — torque-speed curves, overcurrent protection limits, joint position/velocity limits matched to the physical G1 hardware specifications. Domain randomization is applied to mass, friction, sensor noise, communication latency (0–20 ms), and motor response curves.

This two-stage distillation ensures that the emergent behaviors survive the transition from exploration-optimized training to hardware-constrained deployment.

4.4 Stage 4 — Sim-to-Real Transfer

We will follow the established sim-to-real pipeline used by `unitree_rl_gym` and validated by OmniXtreme. Key steps include:

- **Sim-to-Sim validation:** Deploy the trained policy in MuJoCo (independent of Isaac Lab) to verify that behaviors are not simulator-specific artifacts.
- **Domain randomization ranges:** Mass $\pm 20\%$, friction $\pm 40\%$, joint damping $\pm 30\%$, observation noise $\pm 5\%$, action delay 0–20 ms, gravity direction $\pm 3^\circ$.
- **Hardware deployment:** Policy runs on the G1's onboard Jetson Orin via TensorRT-optimized inference at ≤ 10 ms latency (50 Hz control). Communication via `unitree_sdk2_python` over Ethernet.
- **Safety protocol:** Initial tests on a gantry/safety harness. Behavioral kill-switch on torque limits and COM deviation thresholds. Graduated progression from static tool interaction $\rightarrow$ walking + tool $\rightarrow$ full composite tasks.

5. Evaluation Plan & Metrics

Metric	Description	Target
Tool Use Emergence Rate	% of seeds (out of 10) that independently discover tool-use solutions	$\geq 60\%$
Task Success Rate	% of episodes completed successfully after convergence	$\geq 50\%$ (sim)
Exploration Efficiency	Wall-clock time / training steps to first tool-use discovery	$< 500\text{M}$ steps
Behavioral Diversity	Number of distinct tool-use strategies discovered per task (measured via trajectory clustering)	≥ 2 per task

Sim-to-Real Transfer	Success rate of tool-use behaviors on physical G1 hardware	$\geq 30\%$
Ablation Delta	Success rate improvement of RND vs. no-curiosity baseline	$\geq 3\times$ improvement

Table 2. Quantitative evaluation metrics and targets.

Ablation studies:

- RND curiosity vs. ICM curiosity vs. no intrinsic reward (ϵ -greedy exploration only)
- Privileged critic vs. symmetric actor-critic
- Ego depth vision vs. proprioception-only actor (does visual input accelerate tool discovery?)
- Curriculum staging vs. all-tasks-simultaneously training

6. Proposed Timeline

Phase	Duration	Deliverables
Phase 1: Setup	Weeks 1–3	Isaac Lab environment suite, G1 URDF integration, RND module implementation, training infrastructure on cluster
Phase 2: Training & Iteration	Weeks 4–9	Curiosity-driven training across task suite, hyperparameter sweeps, emergence analysis, ablation runs
Phase 3: Distillation	Weeks 10–11	Dagger distillation, residual RL refinement, sim-to-sim validation in MuJoCo
Phase 4: Sim-to-Real	Weeks 12–14	Hardware deployment on G1, real-world evaluation, video documentation
Phase 5: Paper	Weeks 15–16	Paper writing, supplementary video production, code release preparation

Table 3. 16-week project timeline.

7. Required Resources

7.1 Compute

- **Training:** Access to a GPU cluster with ≥ 4 NVIDIA A100 or H100 GPUs for parallel Isaac Lab training (4,096 environments). Purdue's RCAC clusters (Anvil, Gilbreth, Negishi) provide sufficient capacity.
- **Estimated training budget:** ~200–500 GPU-hours for the full task suite including ablations.
- **Inference:** G1's onboard Jetson Orin NX (275 TOPS) for real-time policy deployment at 50 Hz.

7.2 Hardware

- Unitree G1 EDU (43 DOF configuration with Dex3-1 hands) — [existing lab asset / to be procured]
- Safety gantry or overhead tether system for initial real-world testing
- Tool objects: wooden blocks (30×30×30 cm, 1–3 kg), PVC rods (80–120 cm), foam wedges

7.3 Software

- NVIDIA Isaac Lab (Isaac Sim 4.x) with GPU-parallelized RL environments
- unitree_rl_gym (official Unitree RL repository) for G1 URDF and sim-to-real boilerplate
- rsl_rl (PPO implementation) or CleanRL for training
- MuJoCo 3.x for sim-to-sim cross-validation
- TensorRT for onboard inference optimization
- ROS 2 Humble for hardware communication layer

8. Risk Analysis & Mitigation

Risk	Likelihood	Impact	Mitigation
Tool use does not emerge within compute budget	Medium	High	Start with easiest task (Distant Target). Use curriculum learning. Fall back to hybrid approach (light reward shaping + curiosity) if pure curiosity fails.
Sim-to-real gap causes behavior collapse	Medium	Medium	Extensive domain randomization. Residual RL refinement with actuator constraints. Sim-to-sim validation before hardware.
Dex3-1 grasp limitations prevent tool manipulation	Low	High	Design tool objects with graspable handles. Simplify grasp to cylindrical/prismatic shapes. Use magnetic or velcro-assisted grasps as a fallback.
Hardware damage during sim-to-real	Low	High	Safety gantry for all real-world tests. Torque limits enforced in residual RL. Graduated deployment (static → slow motion → full speed).

Table 4. Key risks and mitigation strategies.

9. Expected Contributions

- **First demonstration of emergent tool use on a bipedal humanoid robot** — no prior work has shown curiosity-driven tool discovery on a full-sized humanoid with dexterous hands and locomotion.
- **Open-source task suite** for benchmarking tool-use emergence in humanoid RL, built on Isaac Lab with the G1 URDF — reusable by the community.
- **Empirical analysis of curiosity mechanisms** (RND vs. ICM) in high-DOF embodied settings, providing guidance for future intrinsic-motivation research on humanoids.
- **Sim-to-real pipeline** for complex, multi-phase whole-body behaviors (not just locomotion), extending the OmniXtreme two-stage architecture to manipulation-centric tasks.

10. Key References

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